	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** Academic Year : 2025 – 2026

Presentation on Capstone Project Execution (CPE–316004)

Topic	Ganesh Crane Service Management System	
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Abstract

- The proposed system **“Ganesh Crane Service Management System”** is developed to manage crane booking, crane availability, customer requests, operator details, and service records efficiently.
- Currently, crane booking and management are handled manually, which leads to delays, booking conflicts, and difficulty in tracking services.
- This system provides a digital platform for:
- Managing crane bookings
- Maintaining customer details
- Tracking crane availability
- Assigning operators
- Monitoring booking/payment records

Content

1. Introduction (background of the industry or used based problem / task)
2. Literature Survey
3. Scope of the project
4. Methodology
5. Details of designs, working and processes
6. Results and applications
7. Conclusion and future scope
8. References and Bibliography

1. Introduction

- Crane services are essential in construction, industrial lifting, transport, and heavy equipment movement.
- Traditional crane service businesses manage bookings manually through phone calls/registers which creates issues such as:
- Double booking of cranes
- Difficulty in checking crane availability
- Poor customer record management
- No proper operator assignment system
- Hard to track payments and service history
- Therefore, a computerized crane management system is needed.

2. Literature Review

We studied existing crane rental and booking systems used by local crane service providers.

- Most crane service providers manage bookings manually using registers or phone calls.
- Customer details are recorded in notebooks.
- Crane availability is checked manually.
- Operator assignment is done manually.
- Payment records are maintained separately.

3. Scope of the project

- The system provides an online platform to manage crane services efficiently.
- Users can view available cranes and request crane services.
- Customers can book cranes by selecting date, time, and location.
- The admin can add, update, and delete crane details in the system.
- The system helps manage crane availability and booking records.
- The system maintains a database of cranes, customers, and bookings.

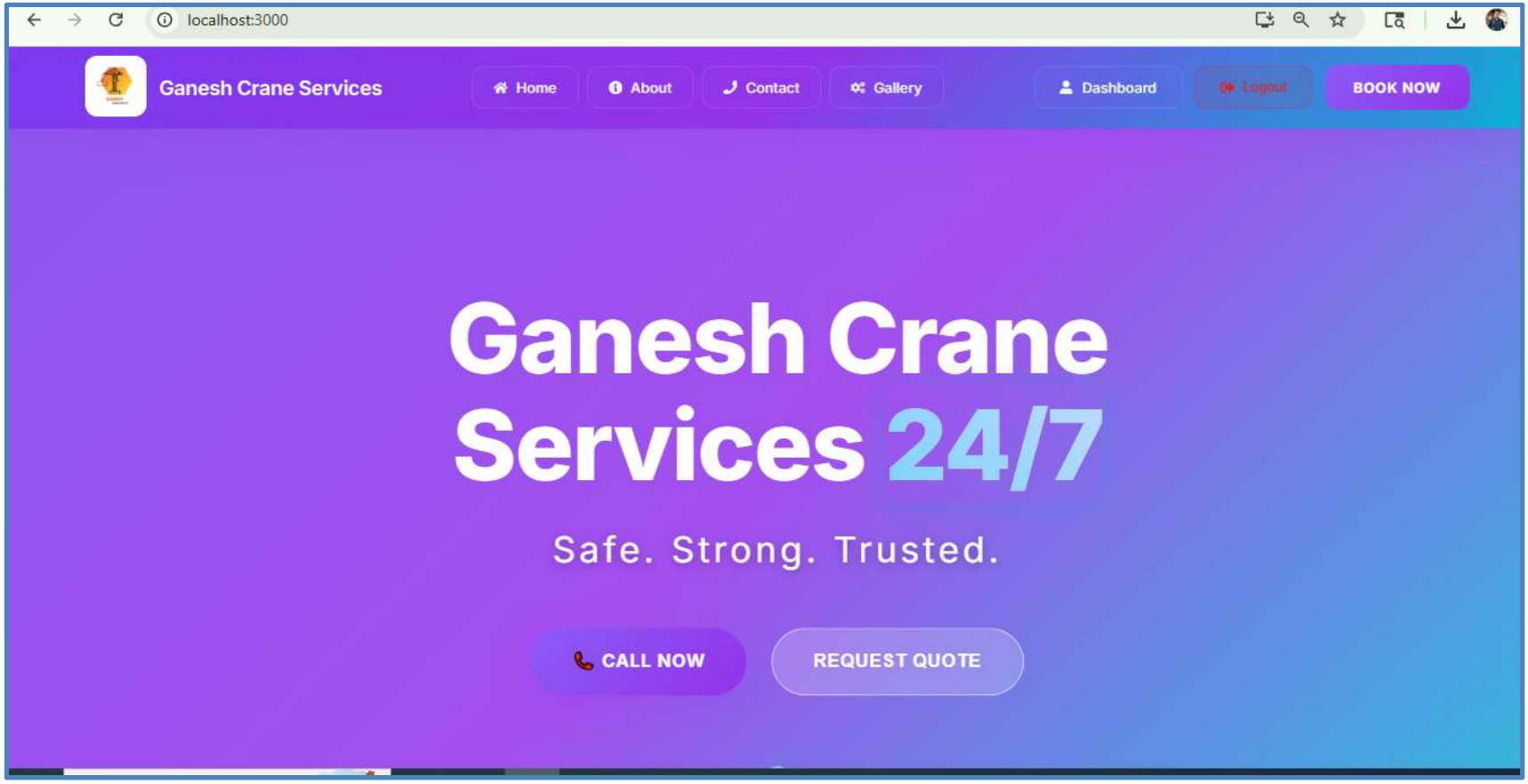
4. Methodology

- Requirement analysis of crane service operations and user needs.
- Designing the system architecture and database structure.
- Development of the web application for crane booking and .
- Integration of user, admin, and crane management modules.
- Deployment of the system for real-time crane service management.

5. Details of designs, working & processes

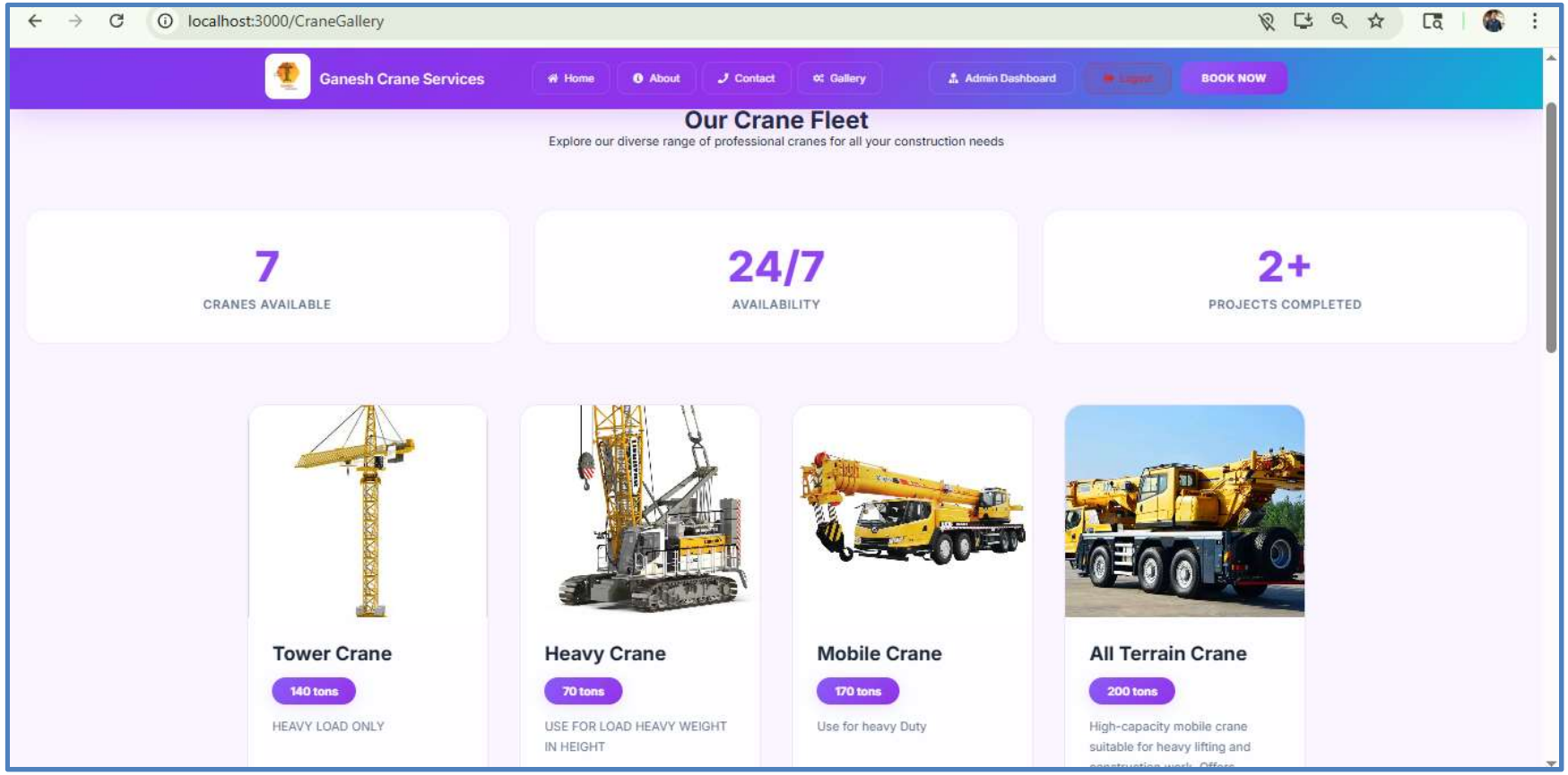
- **1. System Design:** The system is designed that allows users to view available cranes and send booking requests. It includes a user interface and an admin dashboard.
- **2. User Working:** The user registers or logs in to the system, views available cranes, selects the required crane and date, and sends a booking request.
- **3. System Process:** The system checks crane availability, stores the booking request in the database, and updates the booking status.
- **4. Admin Working:** The admin logs into the dashboard, views booking requests, and approves or rejects the booking.
- **5. Final Process:** After admin approval, the booking status is updated and confirmed to the user.

6. Results and Applications



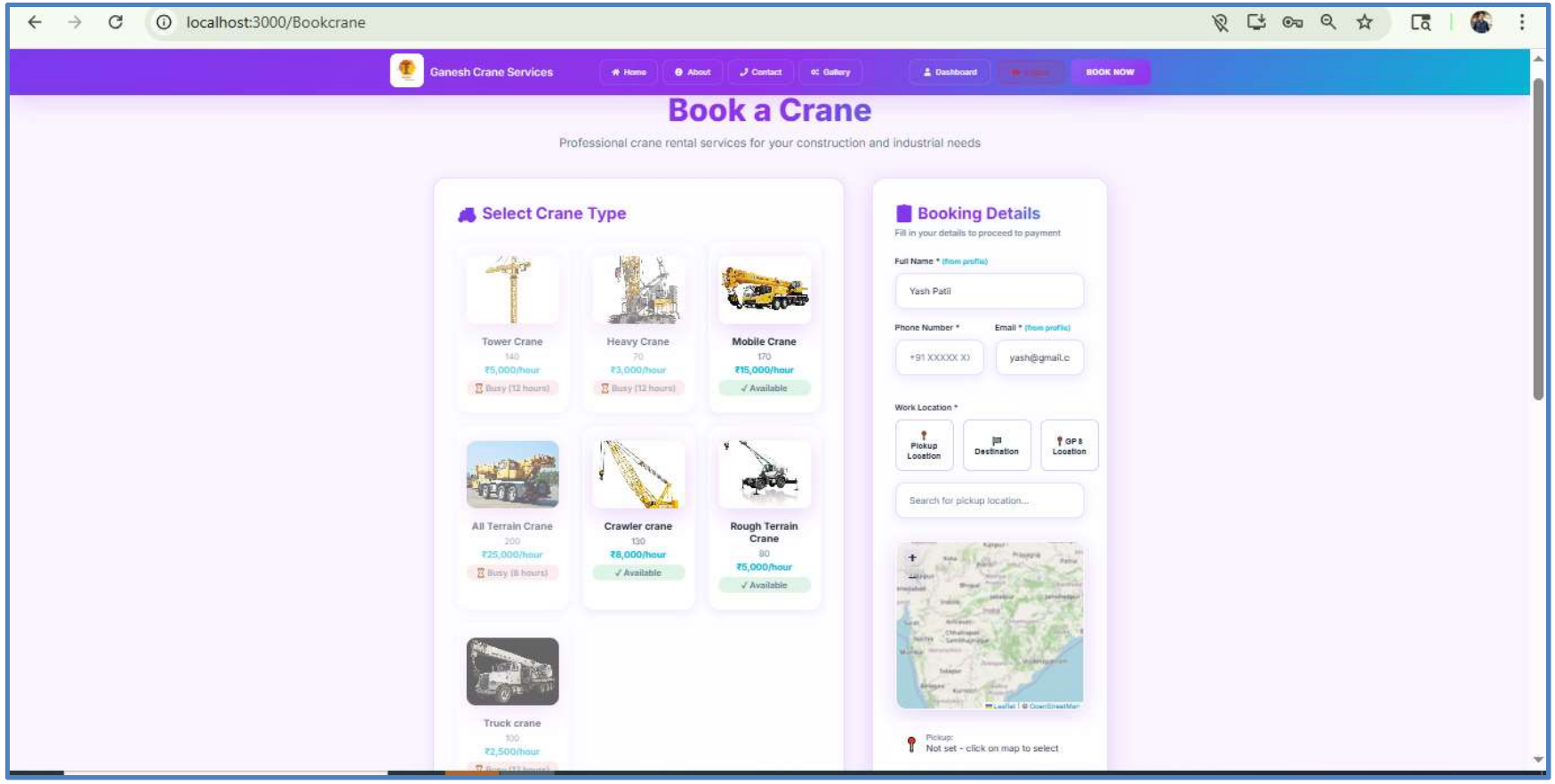
6.1:- Home Page

6. Results and Applications



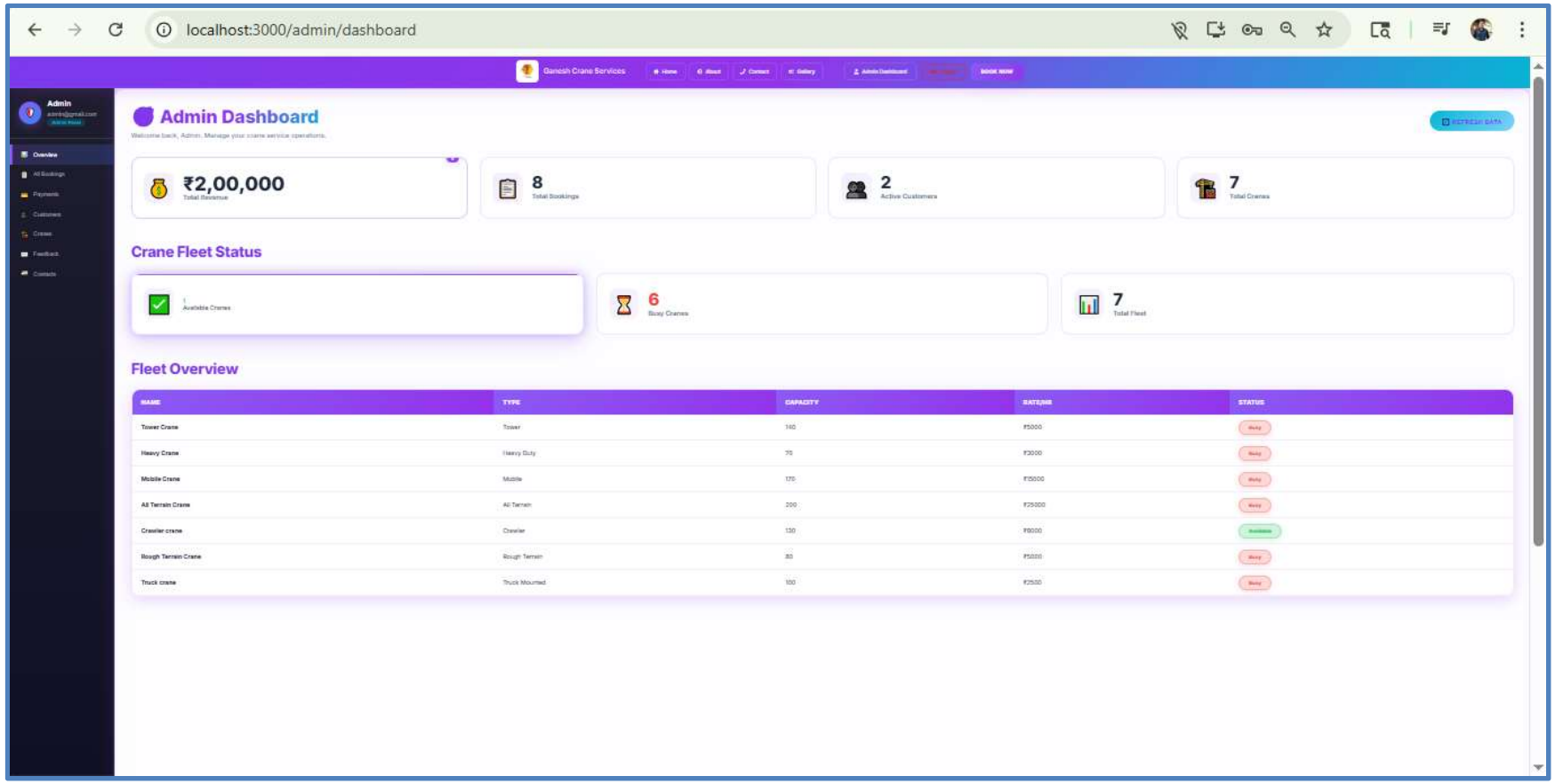
6.2:- Crane Fleet

6. Results and Applications



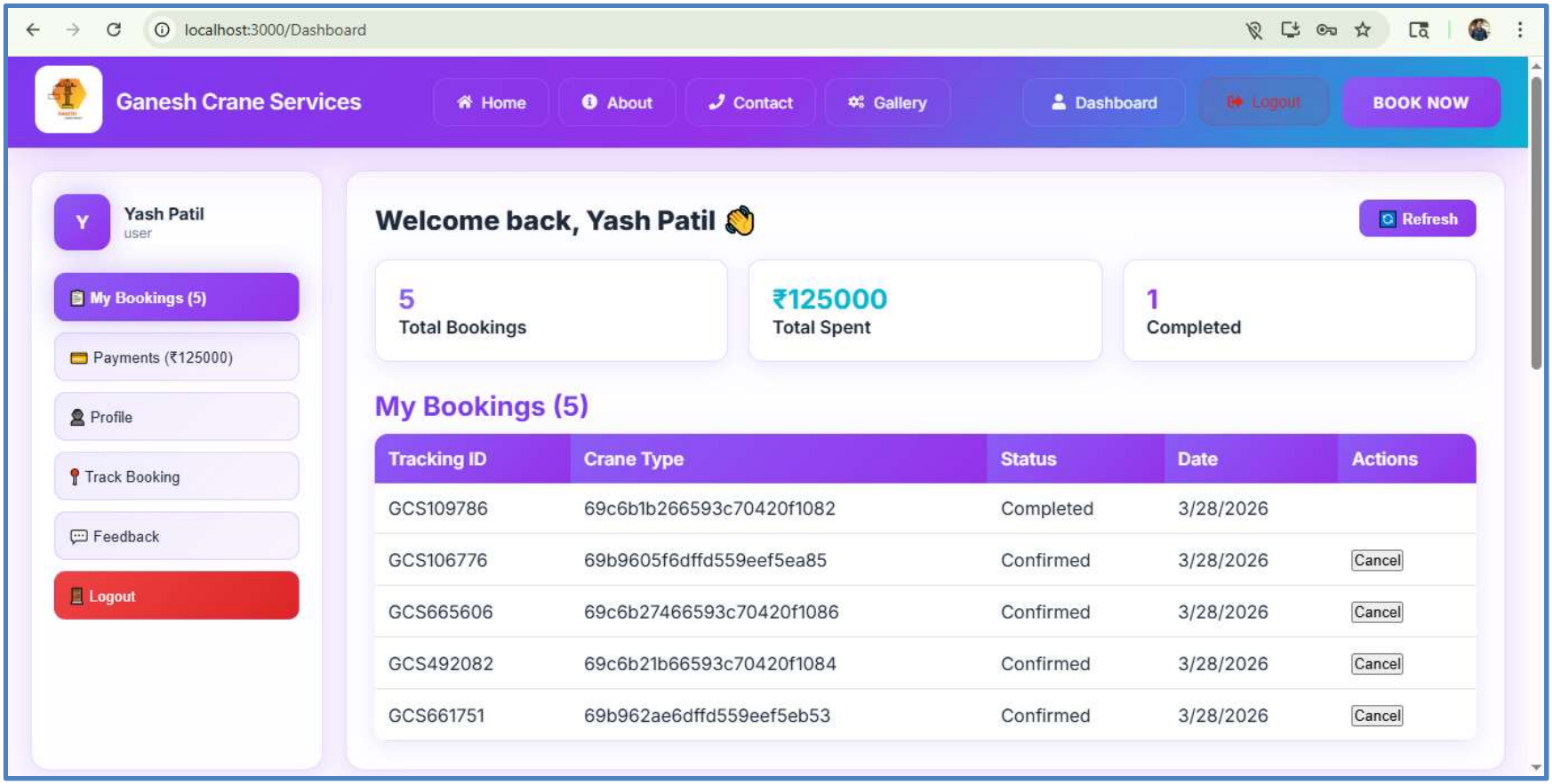
6.3:- Book a Crane

6. Results and Applications



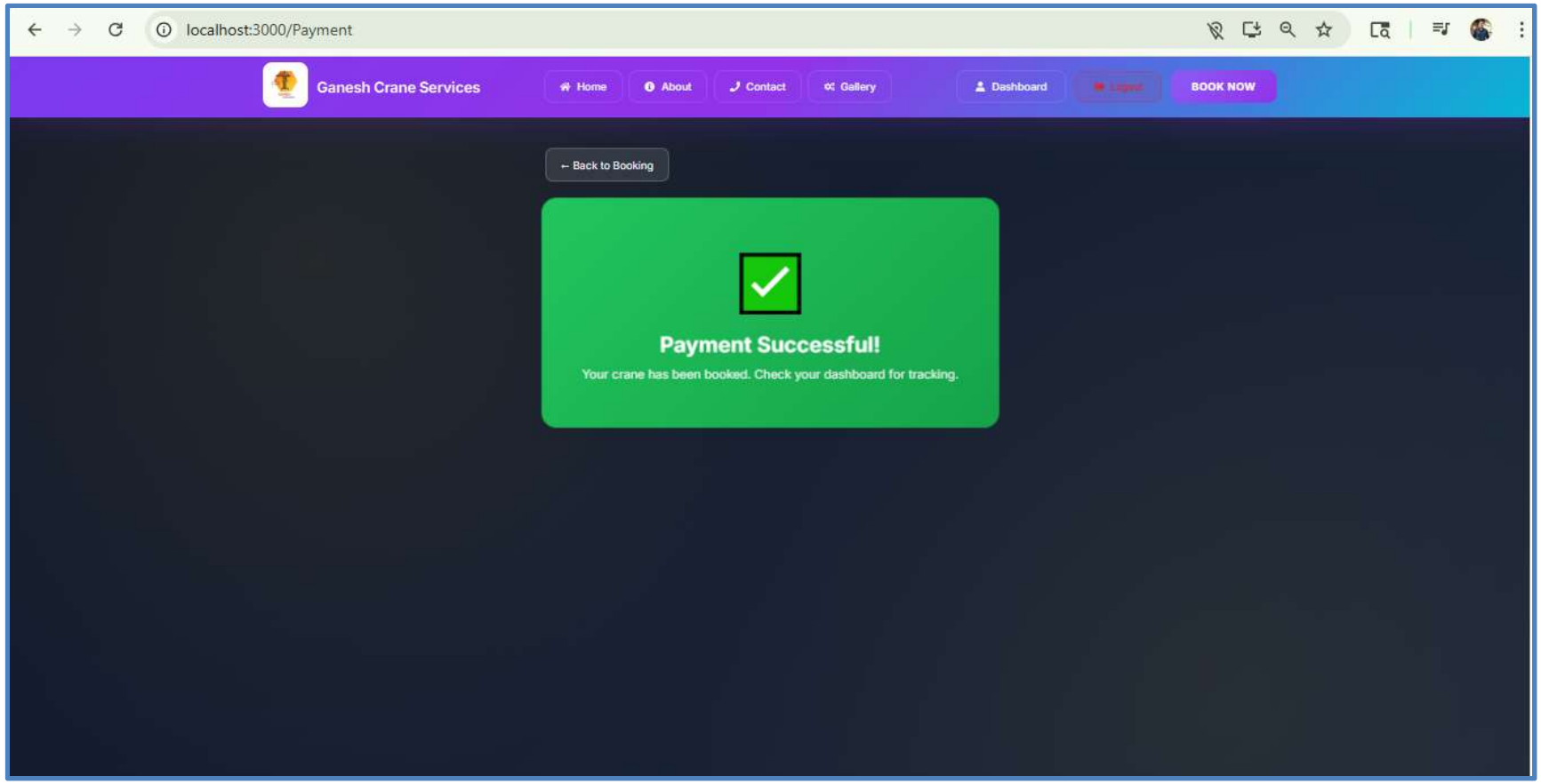
6.4:- Admin Dashboard

6. Results and Applications



6.5:- User Dashboard

6. Results and Applications



6.5:- Payment Successfully

7. Conclusion and future scope

Conclusion:-

The Ganesh Crane Service Management System provides a digital solution for managing crane booking and operations efficiently.

- Benefits:
- Reduces manual work
- Prevents booking conflicts
- Improves customer service
- Maintains proper records
- Generates reports instantly

7. Conclusion and future scope

- **Future Scope:-**
- Mobile Application Development
- GPS Tracking of Cranes
- Online Payment Gateway Integration
- Real-Time Crane Availability Tracking
- SMS/Email Notifications
- AI-based Booking Prediction

8. References and Bibliography

- **References:-**
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- Node JS Official Documentationhttps://nodejs.org/en/docs
- MongoDB Documentation https://www.mongodb.com/docs/
- Online Booking Management System Research Papers https://www.ijraset.com/search?query=online+booking+management+system

- **Bibilography:-**

1. TMEIC. Crane Management System (CMS).
2. Data Scope Systems. Crane Management Software Platform.
3. Manitowoc. Crane STAR Fleet Monitoring System.
4. Agra Cranes & Engineering. Modern Crane Control Systems.
5. Various online articles and research papers on crane monitoring, booking, and equipment management systems.